



MONISHIKA SHARMA

MySQL - db_43xxeqqaq



AI

Run >

```
50 UPDATE StudentEnrollments
51 SET enrollment_date='2024-07-03'
52 WHERE student_name='Ashish' AND course_id='CSE101';
53
54 -- User B: In separate session, attempts to update the same row
55 -- This will be blocked until User A commits
56
57 -- User A commits
58 COMMIT;
59
60 -- Now User B can proceed safely
61 -- After User B commits, data remains consistent; no race conditions
62 SELECT * FROM StudentEnrollments;
63
```

```
mysql> UPDATE StudentEnrollments
SET enrollment_date='2024-07-03'
WHERE student_name='Ashish' AND course_id='CSE101';
ERROR 1205 (HY000): Lock wait timeout exceeded; try restarting transaction
```

enrollment_id	student_name	course_id	enrollment_date
1	Ashish	CSE101	2024-07-03T00:00:00.000Z
2	Smaran	CSE102	2024-07-01T00:00:00.000Z
3	Vaibhav	CSE101	2024-07-01T00:00:00.000Z



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MySQL - db_43xxf64y2



AI

Run >

```
69 START TRANSACTION;
70 UPDATE StudentEnrollments
71 SET enrollment_date = '2024-07-10'
72 WHERE student_id = 1;
73 COMMIT;
74
75 -- Session A continues to see old value
76 SELECT * FROM StudentEnrollments WHERE student_id = 1;
77 COMMIT;
78
79 -- Check final table state after MVCC updates
80
81 SELECT * FROM StudentEnrollments;
82
83
```

```
SELECT * FROM StudentEnrollments;
+-----+-----+-----+-----+
| student_id | student_name | course_id | enrollment_date |
+-----+-----+-----+-----+
| 1 | Ashish | CSE101 | 2024-07-10T00:00:00.000Z |
| 2 | Smaran | CSE102 | 2024-07-03T00:00:00.000Z |
| 3 | Vaibhav | CSE103 | 2024-06-01T00:00:00.000Z |
+-----+-----+-----+-----+
```

student_id	student_name	course_id	enrollment_date
1	Ashish	CSE101	2024-07-10T00:00:00.000Z
2	Smaran	CSE102	2024-07-03T00:00:00.000Z
3	Vaibhav	CSE103	2024-06-01T00:00:00.000Z



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MySQL - db_43xxf64y2



AI

Run >

```
108 START TRANSACTION;
109 UPDATE StudentEnrollments
110 SET enrollment_date = '2024-08-10'
111 WHERE student_id = 1;
112 COMMIT;
113
114 -- Session A still sees old value
115 SELECT * FROM StudentEnrollments WHERE student_id = 1;
116 COMMIT;
117
118 -- Final verification
119 SELECT * FROM StudentEnrollments;
```

```
mysql> START TRANSACTION;
mysql> UPDATE StudentEnrollments
mysql> SET enrollment_date = '2024-08-10'
mysql> WHERE student_id = 1;
mysql> COMMIT;
mysql>
mysql> -- Session A still sees old value
mysql> SELECT * FROM StudentEnrollments WHERE student_id = 1;
mysql> COMMIT;
mysql>
mysql> -- Final verification
mysql> SELECT * FROM StudentEnrollments;
```

student_id	student_name	course_id	enrollment_date
1	Ashish	CSE101	2024-08-10T00:00:00.000Z
2	Smaran	CSE102	2024-07-03T00:00:00.000Z
3	Vaibhav	CSE103	2024-06-01T00:00:00.000Z